Fake News Detection Project Report

This project report presents a Fake News Detector developed using Artificial Intelligence techniques. It aims to help users identify misleading or false information online by analyzing news articles and headlines. The system uses machine learning and natural language processing to classify news as real or fake, addressing the growing problem of misinformation on social media platforms

Problem Statement and Application

Goal

Develop a machine learning program to identify fake news sources by analyzing multiple articles from each source.

Approach

Label sources as fake producers to predict future articles' reliability with high confidence.

Application

Use visibility weights in social media to reduce the prominence of likely fake news stories.

Project Structure and Dataset

Repository Contents

- Images: Diagrams and reports
- Dataset: Train and test CSV files from Kaggle
- Jupyter Notebook: Data analysis and model training
- Flask app: Real-time fake news detection interface
- Pre-trained model and vectorizer files

Dataset Details

Train.csv contains news articles with labels (1 for unreliable, 0 for reliable). Test.csv has similar attributes without labels.

The dataset includes article id, title, author, and text, enabling comprehensive model training and evaluation.

Model Description and Performance

Model Used

The Passive Aggressive Classifier (PAC) is an online learning algorithm suited for binary classification, ideal for fake news detection.

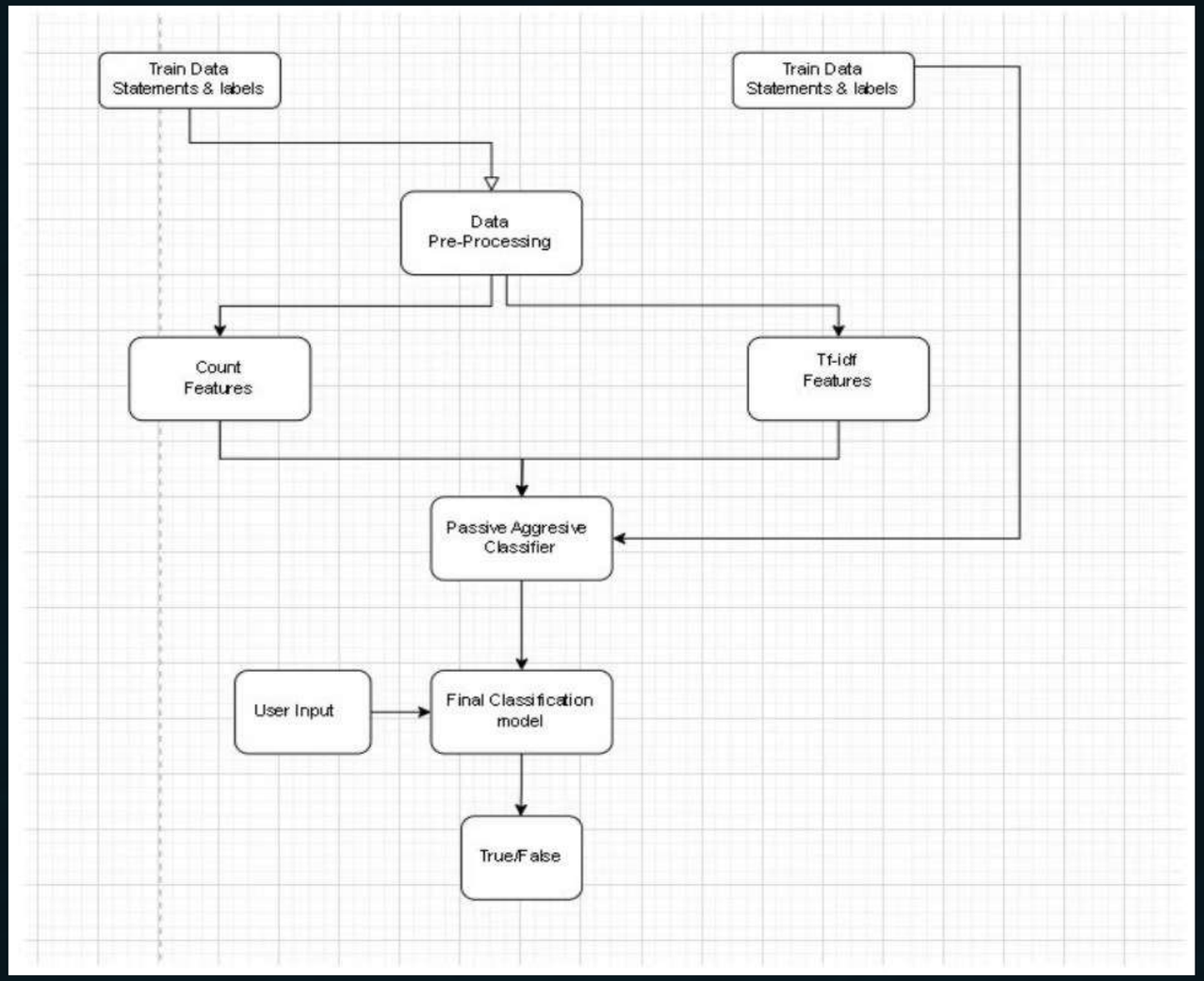
Model Accuracy

PAC achieved a high accuracy of 96%, demonstrating effectiveness in distinguishing real and fake news articles.

Prerequisites

Requires Python 3.7+, dependencies from requirements.txt, and pre-trained model files for prediction.

Key Observations and Insights



Data Quality

High-quality, clean data is critical; noisy or ambiguous data reduces model accuracy significantly.

Vectorization

TF-IDF vectorizer outperformed count-based methods by emphasizing unique, meaningful words.

Model Comparison

Passive Aggressive Classifier balanced speed and accuracy better than Naive Bayes and Logistic Regression.

Deployment

Flask interface improved usability, though real-time prediction requires further optimization for large-scale use.

Proposed Methodology

1

Data Acquisition & Validation

Collect labeled news data from reliable public sources, ensuring dataset authenticity.

2

Preprocessing

Clean text by lowercasing, removing punctuation, stopwords, tokenizing, and stemming for better feature extraction.

3

Feature Extraction

Apply TF-IDF vectorization to highlight important words and reduce noise.

4

Model Training & Evaluation

Train multiple classifiers, select the best based on accuracy, precision, recall, and F1-score metrics.

5

Deployment & Testing

Develop Flask web app for real-time predictions, thoroughly test and debug for reliability.

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Experimental Design and Evaluation

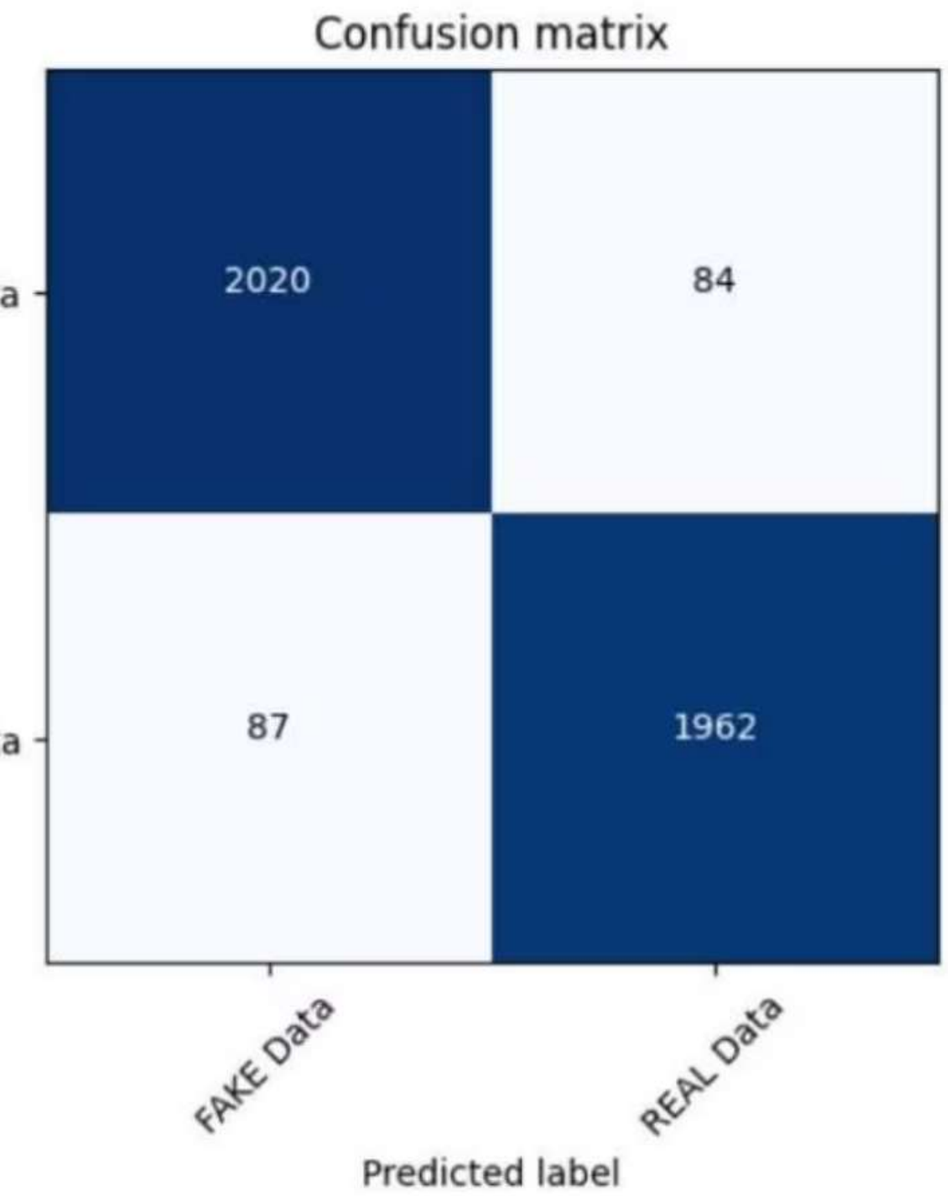
Setup & Tools

- Hardware: Intel i5/i7, 8GB RAM
- Software: Python, Scikit-learn, NLTK, Flask, Pandas
- IDE: Jupyter Notebook, Google Colab

Evaluation Metrics

- Accuracy, Precision, Recall, F1-Score
- Confusion Matrix for detailed error analysis

TF-IDF and Passive Aggressive Classifier yielded the best results with ~94% accuracy.



Conclusion and Future Prospects

This project successfully applied NLP and machine learning to detect fake news with high accuracy, using the Passive Aggressive Classifier and TF-IDF vectorization. The deployment via a Flask web app makes the system accessible to non-technical users, demonstrating real-world applicability.

Future enhancements could include expanding dataset diversity, refining models, and improving deployment scalability to strengthen misinformation detection and support media verification globally.